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**5th Sem / Civil, Brick Tech., Constr. Mgmt.,
Civil Engg. (Spl. Highway Engg.)
Sub : Reinforced Cement Concrete Design & Drawings**

Time : 6 Hrs. M.M. : 150

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The loss of pre-stress is maximum due to (CO11)
a) Friction b) Creep of concrete
c) Shrinkage of concrete d) Elastic shortening
- Q.2 Which method is referred as no deterministic (CO1)
a) WSM b) LSM
c) Both of these d) None of these
- Q.3 The spacing of stirrups near the support is (CO2)
a) Minimum b) Maximum
c) 400 mm d) 350 mm
- Q.4 Design compressive stress is taken as (CO3)
a) 0.50 Fck b) 0.45 Fck
c) 0.40 Fck d) 0.35 Fck
- Q.5 Maximum area of tensile reinforcement shall not exceed (Co4)
a) 1% b) 2%
c) 3% d) 4%
- Q.6 The portion of slab which act with the T beam is called (CO5)
a) Flange b) Web
c) Column d) None of these
- Q.7 The thickness of a two way slab as compared to one way slab is (CO6)
a) Less b) Equal
c) More d) None of these
- Q.8 The minimum Dia. of a longitudinal Steel bar in a column is (CO8)

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|--|----------|----------|--|
| | a) 8 mm | b) 10 mm | |
| | c) 12 mm | d) 16 mm | |
- Q.9 The maximum percentage of longitudinal steel in a column is _____. (CO9)
- | | |
|-------|-------|
| a) 5% | b) 6% |
| c) 7% | d) 8% |
- Q.10 A strand is made of _____ (CO10)
- | | |
|------------|------------|
| a) 5 wires | b) 6 wires |
| c) 7 wires | d) 8 wires |

SECTION-B

- Note:** Objective type questions. All questions are compulsory. (10x1=10)
- Q.11 Factor of safety for concrete is _____. (CO1)
- Q.12 Concrete grade lower than _____ should not be used for RC work. (CO2)
- Q.13 Maximum spacing of shear stirrups is 0.75 D or _____ mm. whichever is less. (CO3)
- Q.14 In a doubly reinforced beam _____ is taken by steel and concrete. (CO4)
- Q.15 In a cantilever beam, the main reinforcement must be embedded into the support for a distance not less than _____ length. (CO5)
- Q.16 The spacing of stirrup near the support is _____. (CO6)
- Q.17 In one way slab bending takes place along the _____ span only. (CO7)
- Q.18 The failure of a short column is by _____. (CO8)
- Q.19 _____ column fails by buckling. (CO9)
- Q.20 Minimum grade of concrete required for pre tensioning is _____. (CO10)

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 What do you mean by HYSD steel? Write some properties of HYSD steel. (CO1)
- Q.22 What are the salient features of the working stress method? (CO2)
- Q.23 Differentiate between singly reinforced beam and doubly reinforced beam (any five) (CO3)

- Q.24 Define nominal shear stress write the formula for it. (CO5)
- Q.25 How can the bond between steel and concrete be increased. (CO5)
- Q.26 Write the assumptions made in the theory of simple bending. (CO6)
- Q.27 What is the purpose of providing distribution reinforcement in one way slab. (CO7)
- Q.28 What is T beam explain in detail. (CO8)
- Q.29 Calculate the ultimate maximum bending moment for a two way simply supported slab of a room 4mx5m the edge of the slab are not held down the live load on the slab is 3.5 Kn/m^2 the slab has a bearing of 230 mm on the supporting walls use (M20) concrete and fe-415 steel also apply the check for depth of slab. (CO9)
- Q.30 An RCC beam 250mm x500mm (effective) has a shear force of 300KN if the tensile steel is 1% find the nominal shear stress in the beam and comment upon shear design assume concrete used is M20 grade. (CO3)
- Q.31 Give comparison between working stress method and limit state method. (Any five) (CO6)
- Q.32 What is the difference between short column and long column. (any five) (CO8)
- Q.33 What is the necessity to provide shear reinforcement. (CO4)
- Q.34 Why the distribution steel is provided on the top on main steel. (CO8)
- Q.35 Write any five difference between pre-tensioning and post tensioning methods. (CO11)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the steps involved in design of limit state method. (CO5)
- Q.37 Find the load carrying capacity of short column of size 400mmx400mm reinforced with 4 bar of 20mm diameter as longitudinal stage. (CO9)

- Q.38 Simply supported RCC slab of a corridor of a school building has a clear span of 3m and supported on beam of 230mm width designed the slab if it is carrying a live load of 6.5N/m use M20 concrete and FE-415 steel. (CO8)

SECTION-E

Note: Long answer type questions. Attempt any two questions out of three questions. (2x25=50)

- Q.39 Draw a suitable scale sectional plan and sectional elevation of a simply supported one way slab with the following data

Size of room = 3m x 6m

Thickness of slab = 150 mm

Bearing of wall = 150 mm

Thickness of wall = 300 mm

Main reinforcement = 12 mm dia. HYSD bar @ 150 mm c/c (with alternate bar bent up at L/7)

Distribution Steel = 10 mm dia. HYSD bar @ 200 MM c/c

- Q.40 Draw to suitable scale cross section of an end connection of a column with a beam from the following data

Column:

Size = 400 mm x 400 mm

Main bar = 8-16 dia

Lateral ties = 8 mm dia. @ 200 mm c/c

Beam

Size = 200 mm x 400 mm (including 120 mm thick slab)

Main bar = 5-20 mm dia. (Two bars bend up at l/7)

Stirrups = 8 mm dia. 2 legged stirrups @ 220 mm c/c

Anchors bar = 2-12 mm dia

- Q.41 Draw a suitable scale the sectional plan and elevation plan of a square column with isolated footing from the following data

Column details :

Size of column = 500 mm x 500 mm

Reinforcement longitudinal bar = 8-20 mm dia.

Transverse ties = 8 mm dia. @ 300 mm c/c (double ties)

Base reinforcement = 12 mm dia. @ 200 MM c/c both ways

Height of column = 3.6 m

Footing details :

Size of footing = 3.0 x 3.0 m

Thickness of footing at column face = 700 mm

Thickness of footing at free end = 400 mm

Depth below G.L. = 1.2 meter

Plinth level above GL = 350 mm

Assume any other missing data